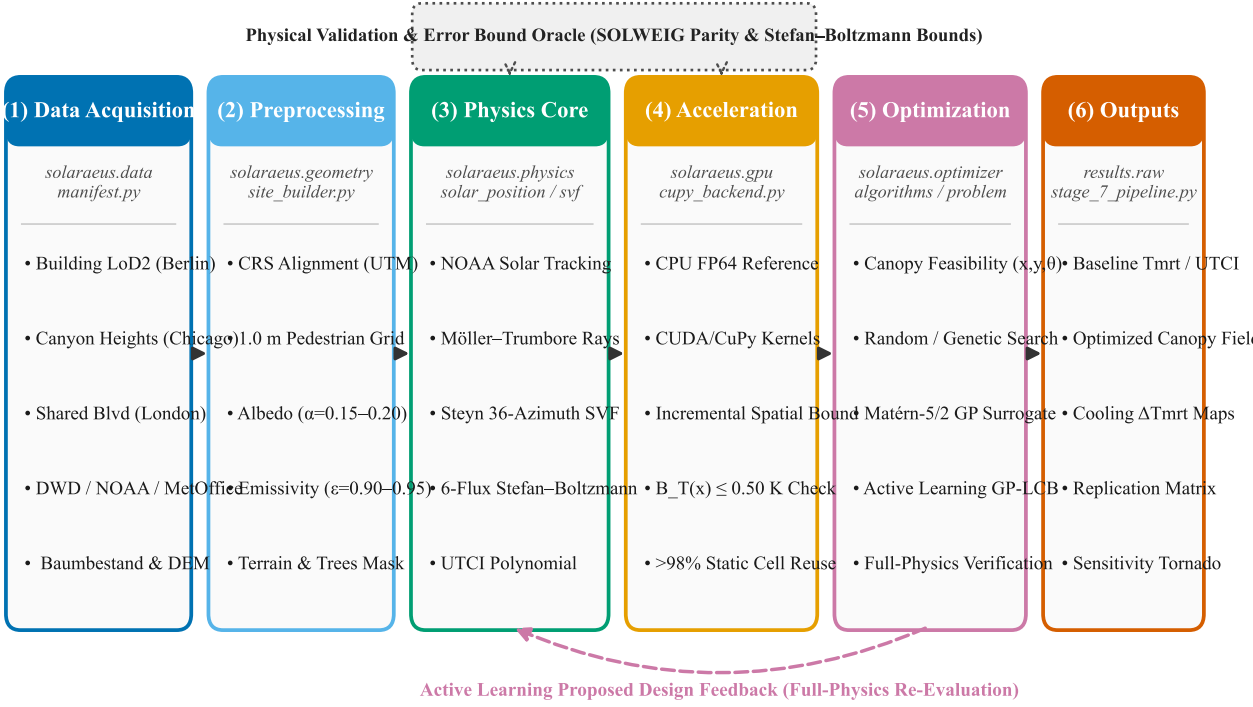

Physical Validation & Error Bound Oracle (SOLWEIG Parity & Stefan–Boltzmann Bounds)
(1) Data Acquisition
solaraeus.data
manifest.py
• Building LoD2 (Berlin)
• Canyon Heights (Chicago)
• Shared Blvd (London)
• DWD / NOAA / Met Office
• Baumbestand & DEM
(2) Preprocessing
solaraeus.geometry
site_builder.py
• CRS Alignment (UTM)
• 1.0 m Pedestrian Grid
• Albedo (α=0.15–0.20)
• Emissivity (ε=0.90–0.95)
• Terrain & Trees Mask
(3) Physics Core
solaraeus.physics
solar_position / svf
• NOAA Solar Tracking
• Möller–Trumbore Rays
• Steyn 36-Azimuth SVF
• 6-Flux Stefan–Boltzmann
• UTCI Polynomial
(4) Acceleration
solaraeus.gpu
cupy_backend.py
• CPU FP64 Reference
• CUDA/CuPy Kernels
• Incremental Spatial Bound
• B_T(x) ≤ 0.50 K Check
• >98% Static Cell Reuse
(5) Optimization
solaraeus.optimizer
algorithms / problem
• Canopy Feasibility (x,y,θ)
• Random / Genetic Search
• Matérn-5/2 GP Surrogate
• Active Learning GP-LCB
• Full-Physics Verification
(6) Outputs
results.raw
stage_7_pipeline.py
• Baseline Tmrt / UTCI
• Optimized Canopy Field
• Cooling ΔTmrt Maps
• Replication Matrix
• Sensitivity Tornado
Active Learning Proposed Design Feedback (Full-Physics Re-Evaluation)